Assignment 7:

1. Ans: axis parallel hyper-rectangles
   Decision trees recursively split the input space using threshold-based rules such as $x_j < c$. These splits are typically **axis-parallel**, producing rectangular regions (hyper-rectangles in higher dimensions).
2. Ans: Option 1 and 2
   Setting a threshold for the decrease in **sum of squared errors (SSE)** prevents insignificant splits, thereby **limiting tree growth**.
   The threshold controls model complexity, so it can **improve or worsen predictive performance** depending on the chosen value.
   The method is **computationally feasible** and is commonly used as a stopping criterion.

3. Ans: Option 2 and 3
   As a decision tree continues to grow:
   Bias decreases → A deeper tree is more flexible and can fit the training data better.
   Variance increases → The deeper tree becomes more sensitive to the particular training data and may **overfit**.
   So, the **bias–variance trade-off** is:
   Tree depth ↑ → Bias ↓, Variance ↑
4. Option:2
   Left branch:

$$response_{left} = \frac{5 + 7 + 9.6}{3} = \frac{21.6}{3} = \boxed{7.2}$$

   Right branch:

$$response_{right} = \frac{8.7 + 9.8 + 10.5 + 11}{4} = \frac{40}{4} = \boxed{10}$$

   After collapsing the node, all **7 data points** are in one node:

$$response_{new} = \frac{21.6 + 40}{7} = \frac{61.6}{7} \approx \boxed{8.8}$$

5. Option D

   Sort the data by feature$_1$:

feature$_1$	**output**
11.7	a

Sort the data by feature$_1$:

feature$_1$	**output**
12.8	a
13.9	a
15.3	a
16.8	b
16.9	b
17.2	a
17.5	b
19.1	b
23.6	b

The candidate split **16.05** lies between:

$$15.3\ (a) \text{ and } 16.8\ (b)$$

So the split gives:

- **Left:** all 4 a values
- **Right:** all 5 b values **plus** the later a at 17.2

Wait—there is an important detail: because **17.2 is a**, this is **not** a perfect split.

Let's calculate the information gain for the candidates. The best split is actually:

17.35

because it separates the data as:

- Left: 5 a + 1 b
- Right: 4 b

This produces the **highest information gain** among the given choices.

6. Using the **weighted Gini index**:

$$Gini(S) = \frac{|S_L|}{|S|} Gini(S_L) + \frac{|S_R|}{|S|} Gini(S_R)$$

Split point	**Weighted Gini**
172.6	**0.2857**
176.35	0.3200

Split point	Weighted Gini
178.45	0.4167
185.4	0.3750

Since **lower Gini impurity is better**, the minimum is:

$$\boxed{0.2857}$$

at **172.6**.

7. Option 1 and 4
   A separate **"Missing" category** is appropriate when the fact that a value is missing is **meaningful information in itself**.

- **Option 1:** Missingness indicates an unusually urgent situation → **meaningful**.
- **Option 4:** Missingness indicates the patient was considered too urgent for the normal process → **meaningful**.
- **Option 2:** Coffee spilled on the records → accidental/random missingness → **not meaningful**.
- **Option 3:** Records were destroyed in a fire → accidental data loss → **not meaningful**.